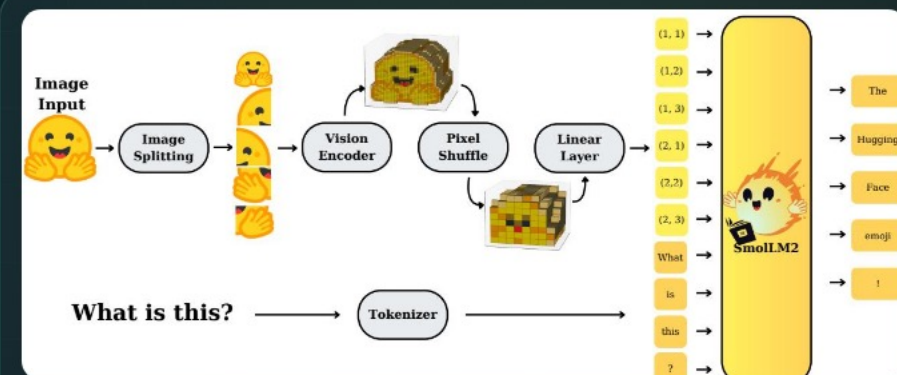


HUGGING FACE × STANFORD · EFFICIENT VLMS

SmoIVLM: *small* multimodal models, rebalanced

A compact VLM family—256M, 500M, and 2.2B parameters—built for resource-efficient inference on mobile and edge devices.



Source figure: SmoIVLM architecture pipeline

Large VLMs are powerful— but impractical at the edge.

Mobile and low-power devices need multimodal reasoning without cloud dependency or workstation-class memory.

27.7GB

Molmo-A1B-7B RAM
requirement

0.8GB

SmoVLM-256M RAM at
batch=1

300×

SmoVLM-256M surpasses
larger Idefics-80B

- ▶ **Constraint:** inference memory determines deployability on smartphones and edge hardware.
- ▶ **Opportunity:** architectural efficiency can beat raw parameter scale.
- ▶ **Goal:** retain competitive multimodal accuracy under strict memory budgets.

Image splitting → SigLIP encoder → pixel shuffle → SmoLM2.

01

Sub-images

Images are split before visual encoding.

02

SigLIP vision encoder

Compact encoder maps patches to visual features.

03

Pixel shuffle

Space-to-depth compression lowers visual tokens.

04

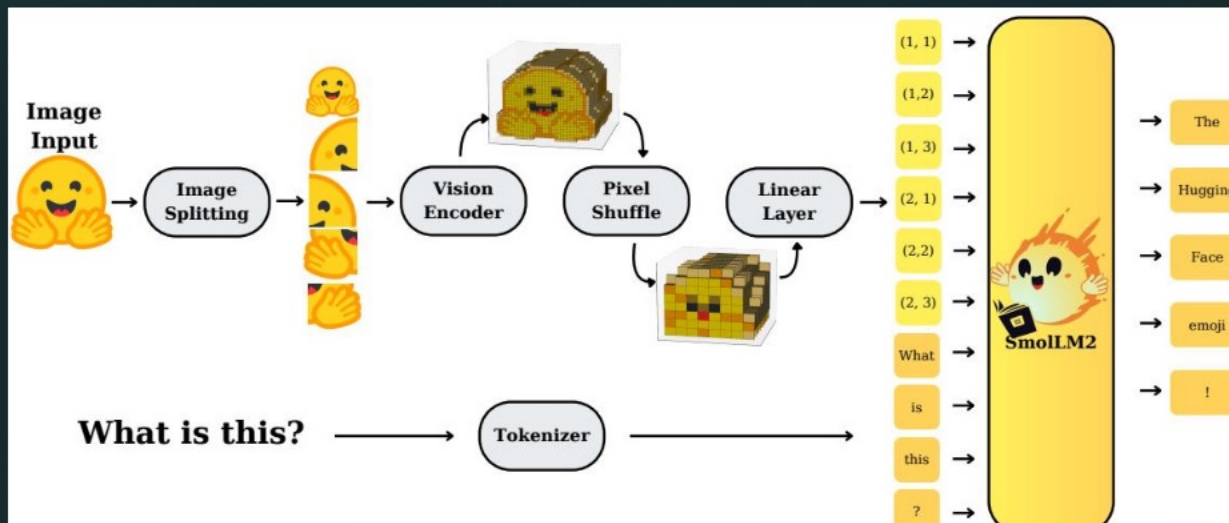
Linear projection

Features enter the language input space.

05

SmoLM2 backbone

Visual and text embeddings are concatenated.



FINDING 1 · COMPUTE ALLOCATION

For compact LMs, the smaller vision encoder wins.

A 93M SigLIP-B/16 paired with a 135M LM outperforms a 428M SigLIP-SO400M with the same LM; the larger encoder only justifies its 10% parameter overhead at 1.7B LM scale.

93M SigLIP-B/16 + 135M LM

outperforms

balanced

428M SigLIP-SO400M +
135M LM

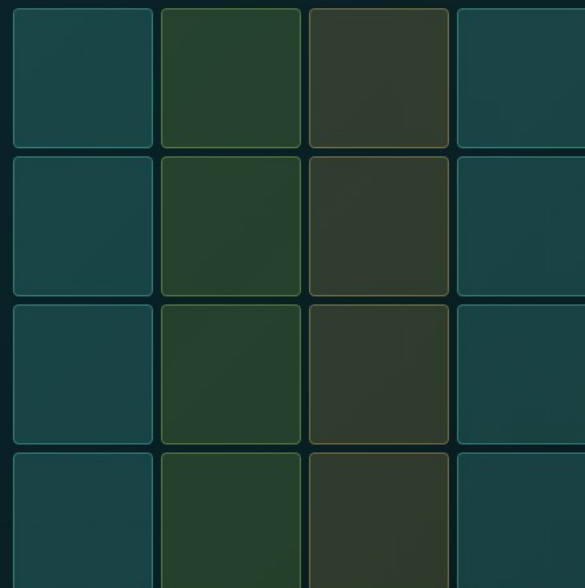
larger encoder

underperforms

Design rule: allocate parameters to match the language model scale, not to maximize visual encoder size.

Pixel shuffle at $r=4$ cuts visual tokens 16x.

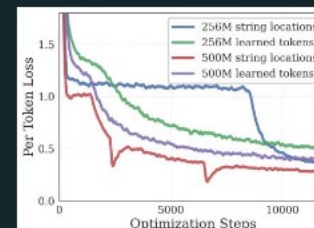
- ▶ **Mechanism:** rearrange spatial features into channels.
- ▶ **Reduction:** visual token count decreases by r^2 .
- ▶ **Small models:** benefit from aggressive $r=4$ compression.
- ▶ **Larger models:** use $r=2$ in the paper design.



16 spatial tokens

1 token · 16 channels

$$r = 4 \Rightarrow r^2 = 16\times \text{ fewer tokens}$$



FINDING 2 · CONTEXT SCALING

Extending context from 2K to 16K unlocks higher-resolution inputs.

SmoIVLM increases RoPE base from 10K to 273K and fine-tunes on mixed long/short-context data.

baseline context

2K

short context
constrains visual token
budget

smaller variants

8K

supports compact
models with expanded
visual input

2.2B variant

16K

largest SmoIVLM uses
extended context

Long context and token compression work together: fewer visual tokens make attention overhead manageable.

Competitive scores at 0.8GB–4.9GB RAM.

Capability	Benchmark	256M	500M	2.2B	Best Efficient OS
Single-image	OCRBench	52.6%	61.0%	72.9%	54.7% Molmo-A1B-7B
Single-image	AI2D	46.4%	59.2%	70.0%	71.0% Molmo-A1B-7B
Single-image	ChartQA	55.6%	62.8%	68.7%	48.0% Molmo-A1B-7B
Single-image	TextVQA	50.2%	60.2%	73.0%	61.5% Molmo-A1B-7B
Single-image	DocVQA	58.3%	70.5%	80.0%	77.7% Molmo-A1B-7B
Single-image	ScienceQA	73.8%	80.0%	89.6%	87.5% Molmo-A1B-7B
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% Molmo-A1B-7B
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% Molmo-A1B-7B
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% Molmo-A1B-7B
Video	Video-MME	33.7%	42.2%	52.1%	45.0% InternVL2-2B
Video	MLVU	40.6%	47.3%	55.2%	48.2% InternVL2-2B
Average	All benchmarks	44.0%	51.0%	59.8%	—

0.8GB

SmoIVLM-256M

1.2GB

SmoIVLM-500M

4.9GB

SmoIVLM-2.2B

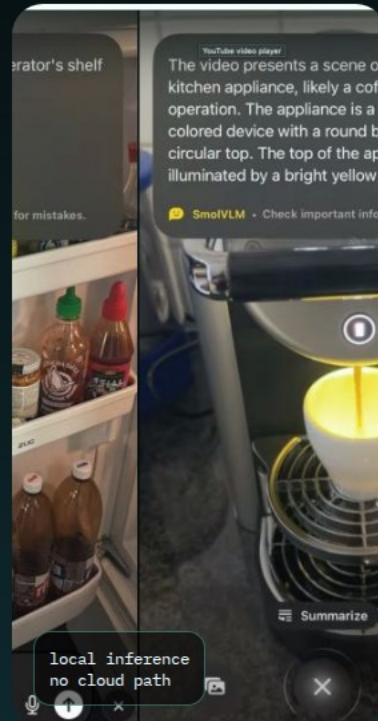
27.7GB

Molmo-A1B-7B

ON-DEVICE DEPLOYMENT

Real-time local inference via HuggingSnap.

- ▶ **iOS app:** photos and videos processed entirely locally.
- ▶ **Pro M1 Max 64GB:** 256M reaches up to ~80 tokens/sec.
- ▶ **Batch size 64:** nearly 1000 tokens/sec on high-end devices.
- ▶ **MacBook Air M3 16GB:** 256M produces ~55 tokens/sec.



Pro M1 Max 64GB

~80 tok/s

MacBook Air M3 16GB

~55 tok/s

batch=64 high-end

~1000 tok/s

Throughput scales roughly linearly with batch size for the smallest model.

Efficient multimodal models are designed, not merely downsized.

01

Balance encoder-LM compute

Compact VLMs benefit from smaller vision encoders matched to LM scale.

02

Extend context length

RoPE base 10K – 273K plus mixed long/short fine-tuning supports 8K–16K context.

03

Compress visual tokens aggressively

Pixel shuffle $r=4$ gives 16× token reduction for small models.

04

Target memory first

Sub-1GB inference is achievable: 256M runs at 0.8GB RAM.

05

Open the full stack

Weights, datasets, code, and HuggingSnap mobile app are fully open-sourced.

Implication: on-device AI research should optimize allocation, token budget, and context—not just parameter count.